Printed Pages:02 **Sub Code:NIC-044**

Paper Id: 131307 **Roll No.** | | | | | | | | | | |

B TECH

(SEM-VIII) THEORY EXAMINATION 2018-19

DIGITAL SYSTEM DESIGN USING VHDL

Time: 3 Hours ***Total Marks: 100***

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

a. Define: Signals.
b. Define: Constant.
c. Listed the applications of generics.
d. What are the Binding alternatives?
e. Listed the various design flow of VHDL coding.
f. Differentiate between transport and Inertial delay models.
g. Write VHDL code 2: 1 Multiplexer.
h. Explain concurrent statement with respect to VHDL.
i. What are different levels of abstractions of digital design?
j. Difference between VHDL functions and procedures.

SECTION B

2. Attempt any *three* of the following: 10 x 3 = 30

a. (i) Explain Nine-Valued logic System for IEEE-1164 standard logic.
(ii)Write VHDL statement for synchronous and asynchronous T-flip-flop.
b. Discuss Sequential modeling and attributes.
c. What is Guarded signal Assignment? Explain with example.
d. What are the Multiple concurrent drivers? Explain different resolving methods.
e. Write short note on the followings with suitable diagram:
(i). FPGA. (ii) CPLD.

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SECTION C

3. Attempt any *one* part of the following: 10 x 1 = 10

(a) Using single bit adder, write a VHDL code for 4 bit adder.
(b) Write short notes on linked state machine.

4. Attempt any *one* part of the following: 10 x 1 = 10

(a) Design a behavioral model for 4x4 binary multiplier using state machine chart.
(b) Explain the application and the use of status word register of UART transmitter.

5. Attempt any *one* part of the following: 10 x = 10

(a) (i)Give the syntax for wait and loop statements with examples for each.
(ii)Write a VHDL code for synthesis of sequential case statement.
(b) Write a short note on XILINX-4000 FPGA

6. Attempt any *one* part of the following: 10 x 1 = 10

(a) Write behavioral VHDL code for RAM.
(b) Write and explain a state diagram for 010 sequence detector. Write VHDL code for the same.

7. Attempt any *one* part of the following: 10 x 1 = 10

(a) What do you mean by Testing? What are different issues related to design the test?
(b) Explain generic BIST scheme.

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